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(54) **FULL-UNIT LIFTING METHOD FOR
QUAYSIDE CONTAINER CRANE**

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(57) **ABSTRACT**

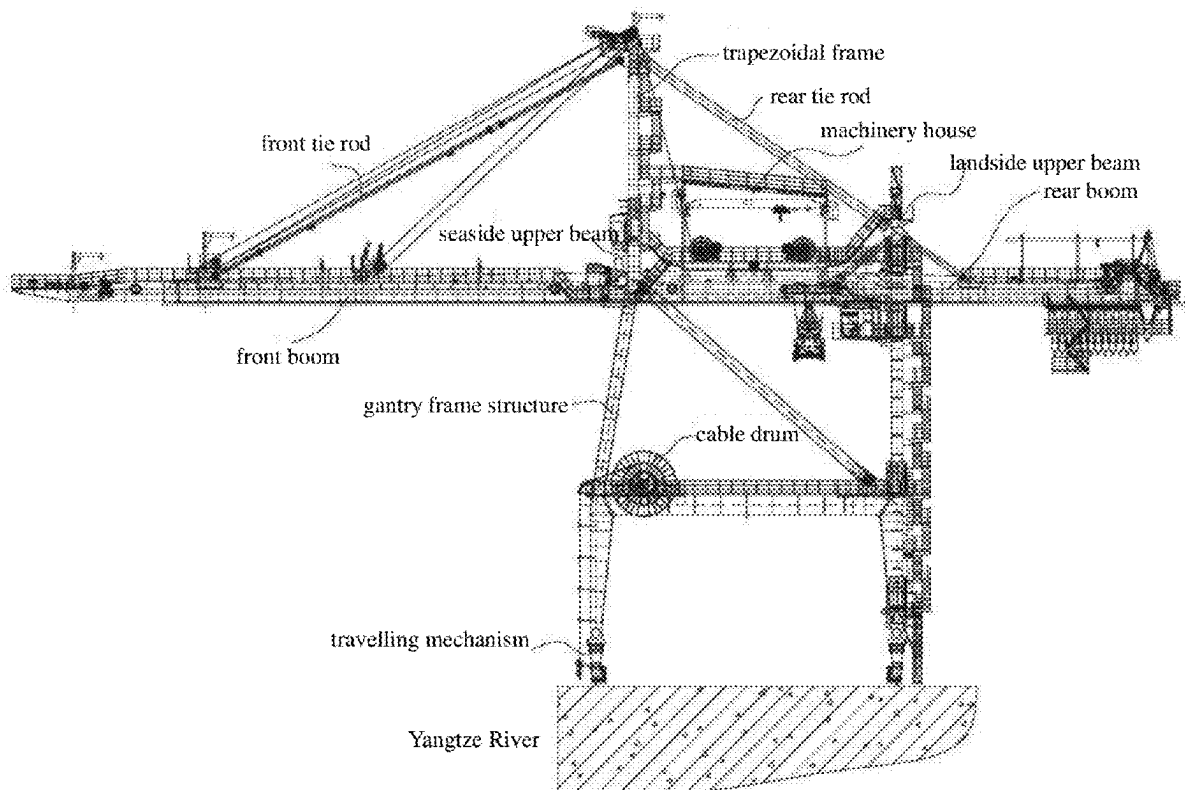
A full-unit lifting method for a quayside container crane includes calculating an overall center position of gravity of the quayside container crane full-unit; estimating a lifting height and a floating amplitude required by a main hook of a floating crane according to a water level differential at a dock and a height of the quayside container crane full-unit; selecting the floating crane and obtaining detailed parameters thereof according to basic lifting conditions of the floating crane; selecting a type of the main hook according to a weight of the quayside container crane full-unit, checking whether an interference between the boom of the floating crane and the front boom occurs, and determining an inclination angle of the floating crane during lifting; calculating a load on the lifting wire rope and selecting the lifting wire rope and a shackle; making preparations before lifting; and performing lifting at the dock.

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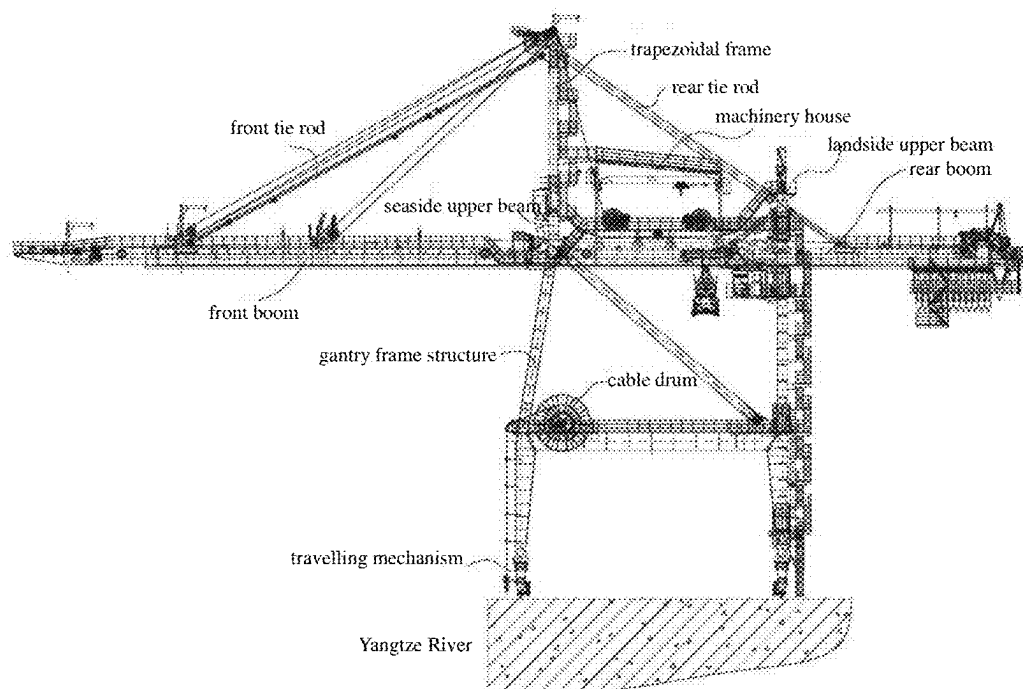


FIG. 1

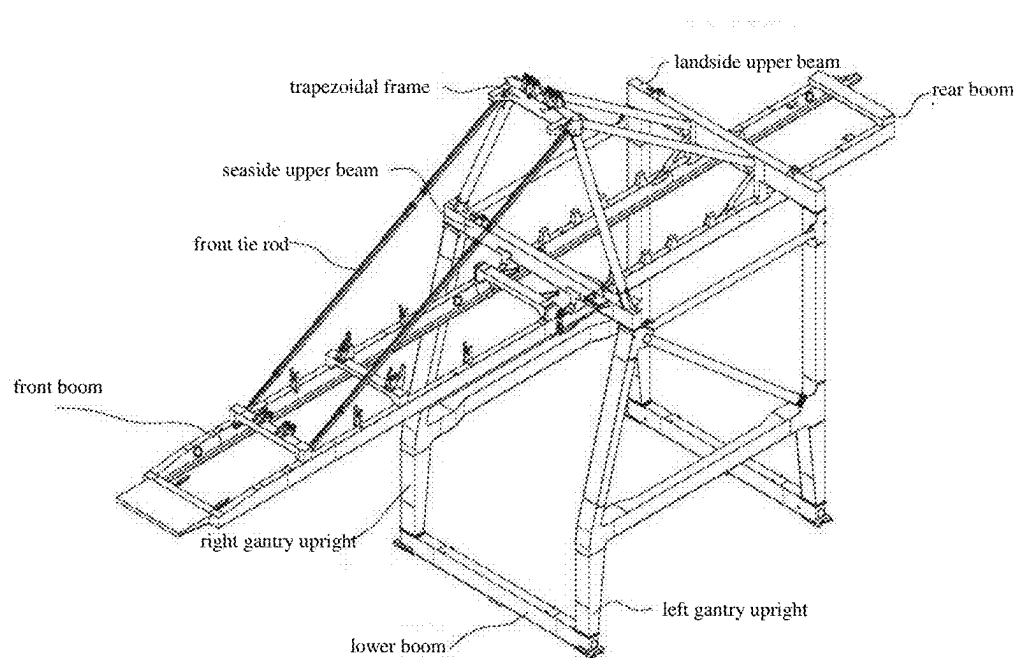


FIG. 2

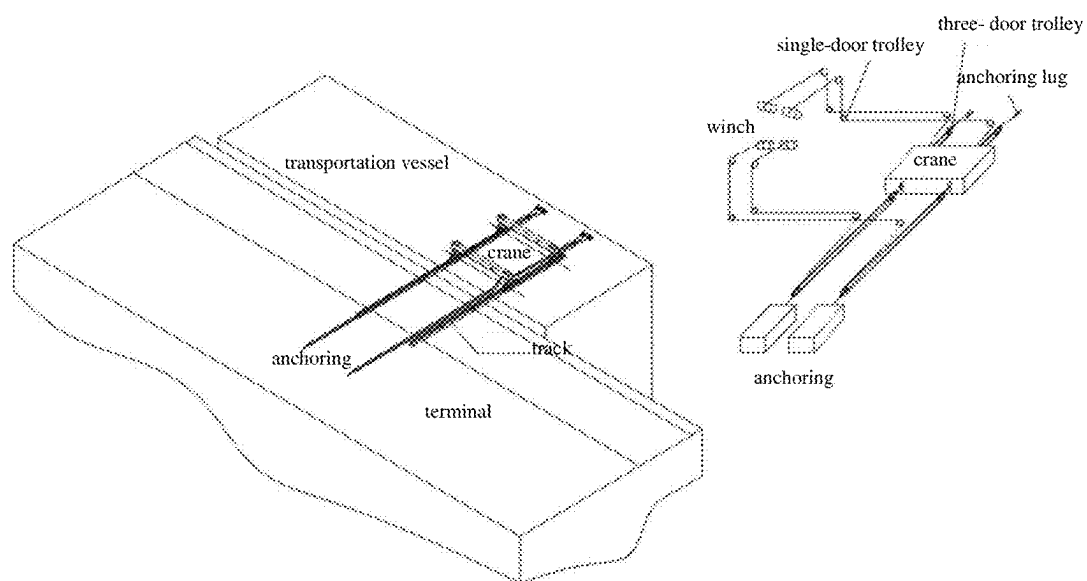


FIG. 3

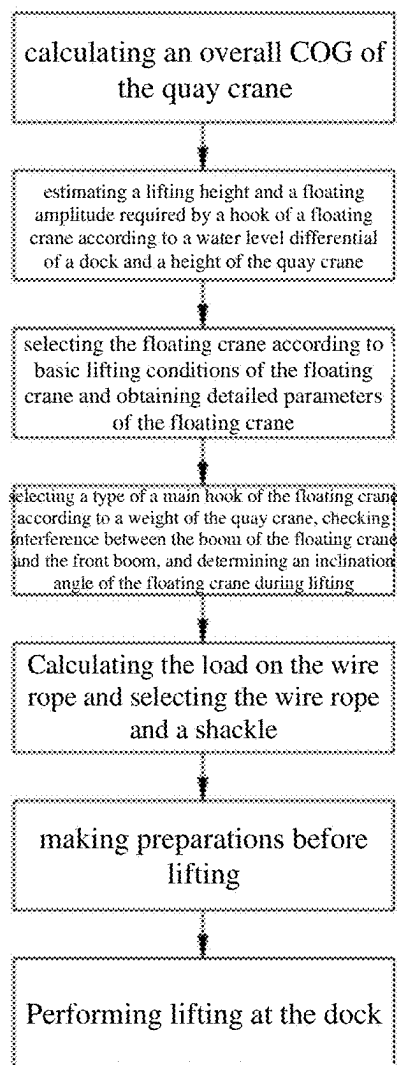


FIG. 4

4.05t quay crane COG table									
number	component	weight(t)							
	seaside trapezoidal frame								
1	seaside upper beam	8.561	4551.80	1.180	3055.80	1.560	3875.80		
2	left gantry upright	20.996	1021.82	0	0	0	46.63	2895.86878	
3	right gantry upright	19.127	0	0	0	0	34.93	1917.574124	
4	landside upper beam	60.9417	12.89	424.289513	0.885	424.2895137	17.182	11128.35733	
5	gantry lower beam	26.5093	0	0	0	0	30.998	1889.72982	
6	gantry vertical ladder	18.208	0	0	0	0	1.85	189.236242	
gantry total	gantry structure	1.7574	12.89	22.702862	0	1.265	17.182	14.286262	
7	rear boom	280.2271	12.82073277	2.262188187	0.000631288	0	12071112	28.31348	27827.73134
8	front tie rod	72.249	22.872	1638.24698	0.0	0	36.1249	21.112	20949.11189
9	rear tie rod	11.9688	11.175	125.484992	0	0	41.849	638.879391	
10	trolley track (front)	12.5488	15.127	186.90280	0	0	41.3	2827.252684	
11	trolley track (rear)	9.9328	28.4785	183.981344	0	0	31.014	1146.133743	
12	front boom	1.9738	21.88	181.73907	0	0	31.983	1884.89161	
13	towing track (front)	14.07	38.548	1896.71854	0	0	17.1	5827.22481	
14	towing track (rear)	1.338833028	18.984	81.5151404	0	0	30.246	835.469388	
15	machinery house chassis	1.681876976	28.851	78.3634919	0	0	38.233	1179.371632	
16	hinge point assembly	47.7794	12.692	861.818209	1	47.7794	31.298	20892.99249	
17	electrical opening cable	2.1912	1	2.1312	0	0	32.828	58.337172	
metal structure total	total of metal structures	2.2624	0.000	32.106957	1.0	1.9086	28.534	782.139732	
18	travelling mechanism	470.5125	6.8901256	2021.11122	0.00022316	19.4292885	22.6282	22281.07522	
19	anti-wind device	77.39	0	0	0	0	1.07	1206.2984	
20	cant anchoring device	8.40	0	0	0	0	0.43	12.0188	
21	elevator	1.2652	0	0	0.0	0.82884	1.7	181.28898	
22	gantry ladder landing	3.8888	21.28	71.218926	11.8	38.28222	8	587.588264	
23	front boom ladder landing	12.6682	22.5	122.42338	11	128.2832	17.38	1684.81168	
24	rear boom ladder landing	8.6273	28.492	213.694341	0	0	31.943	1315.922448	
25	seaside trapezoidal frame ladder landing	22.18	17.843	413.288256	0.0	0	12.09	1282.92201	
26	seaside upper beam landing	1.7884	1.782	38.873771	0.0	0	1.066	1886.87112	
27	landside upper beam landing	1.7884	1.121	7.2030246	0	0	13.8	203.690514	
28	machinery house ladder landing	6.3396	23.732	66.1351388	0	0	37.577	1458.443461	
29	cable towing maintain platform	4.822	12.892	62.9134	0	0	38.595	5897.61991	
30	roof platform of machinery house	6.328	32.715	281.989379	0.48	81.86128	27.888	1834.698882	
ladder platform total	total of ladder landings	8.839	71.26	19.880162	0	39.126	491.2541096		
31	total of travelling trolley	25.2135	15.4959844	715.953679	1.21718881	68.8487	28.2788	1346.18717	
32	total of machinery house	1.6818	0	0	0	0	31.288	1239.72882	
33	fully room	45.5738	11.45	261.818272	1	45.5738	17.5	14908.11126	
34	outdoor travelling device	1.28	0	0	0.0	0.8129	1.4	18	
35	cable drum	3.222	0	11.769	0.0	0.2646	34.788	207.382484	
36	cable towing device	2.9592	1.0	30.94496	11.1	36.72368	15.3	686.294813	
37	main lifting mechanism	6.9529	48.3	28.46187	0.0	0	12.088	28.142	1129.728272
38	trolley traction mechanism	24.6951	12.882	121.484729	0	0	34.911	14936.39202	
39	pitching mechanism	7.7789	12.554	86.782844	0.8	8.8298	12.3	2946.248282	
40	emergency mechanism	18.063	0.063	123.710281	0.0	13.5747	34.373	4692.746722	
41	lifting winding system	0.1812	18.12	8.304310	0	1.8254	38.233	374.691588	
42	trolley winding system	8.5193	1.722	15.9857176	0	0	31.782	335.858122	
43	pitching winding system	3.121	0.0	13.4112	0	0	31.782	463.659348	
44	safety hook	8.99	10.172	98.14527	0	0	41.512	3988.075897	
45	pitch operation room	4.2214	1.748	8.3226264	0	0	31.788	447.951987	
46	tilting mechanism	1.1121	1.514	6.9514936	0	0	36.596	71.1268722	
47	bracket trolley winding system	1.0963	44.050	180.3069607	0.0	0.64910	32.576	5791.867082	
48	hanger protection device	8.1124	77.7	168.17248	0	0	31.2	3375.911778	
49	lubrication system	13.6429	41.682	393.395886	0	0	38.888	21245.38312	
50	hangers and shelves	1	0.03	0.03	0	0	0.03	0.06	
51	electric	12.2	0	0	0	0	27.234	1471.4210	
52	load	18	18.738	128.30	1	18	34.981	1351.38888	
pitch part PG2=BL (front boom and accessories)		80.0	0	1417.0	0	0	39.432	8428.4172	
fixed part PG1=DL-BL		108.123322	41.2788826	2322.14872	0	0	21.8888	1428.51112	
moving part PG3=TL		876.464667	13.7341164	9821.18518	0.35432818	204.3076888	24.3387	2318.014812	
lifting load (on-load)		17.6016	0	0	0	0	31.489	1333.72824	
lifting load (no-load)		28.2	0	1543.0	0	0	46.18818	16395.6285	
on-load		126.2	89	327	0	0	27.272	13271.638	
off machine	no hoist	459.9833	7.2571165	3926.4241	0.208611143	250.8938932	20.1986	91861.89757	
	no load	803.2938	7.89945779	6892.8611	0.114351135	133.3872885	31.7519	128129.3221	
		813.4032	7.27718937	5828.8941	0.10718963	220.3870887	31.288	128317.9282	

FIG. 5

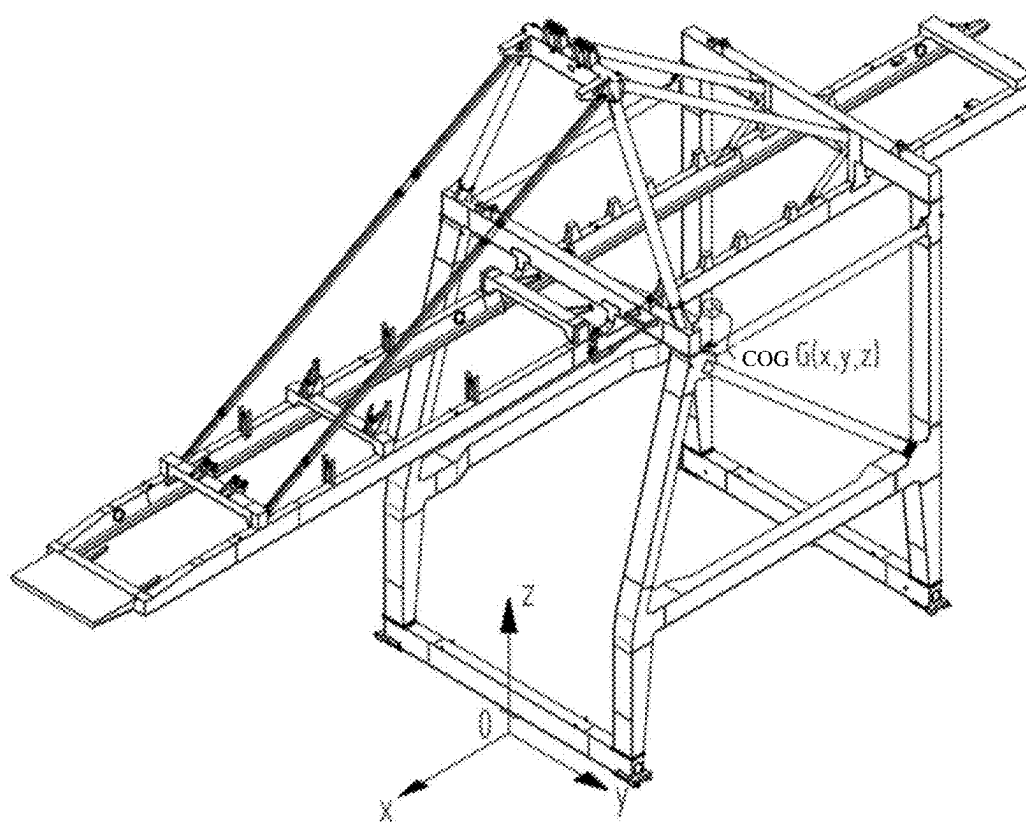


FIG. 6

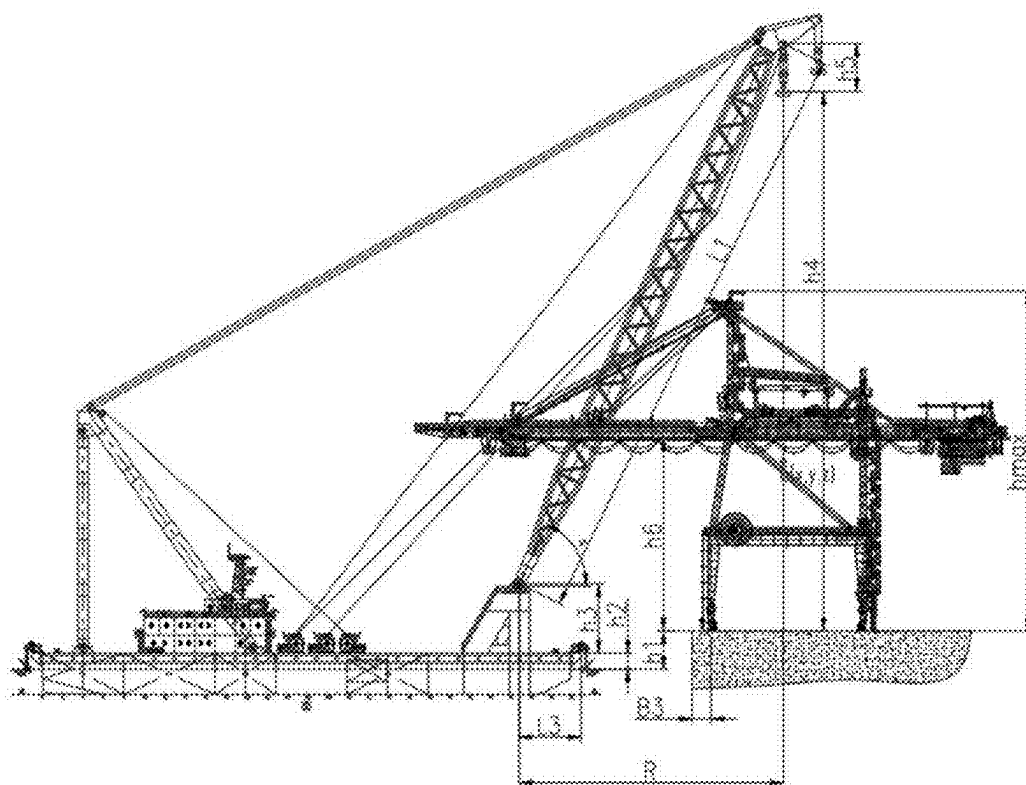


FIG. 7

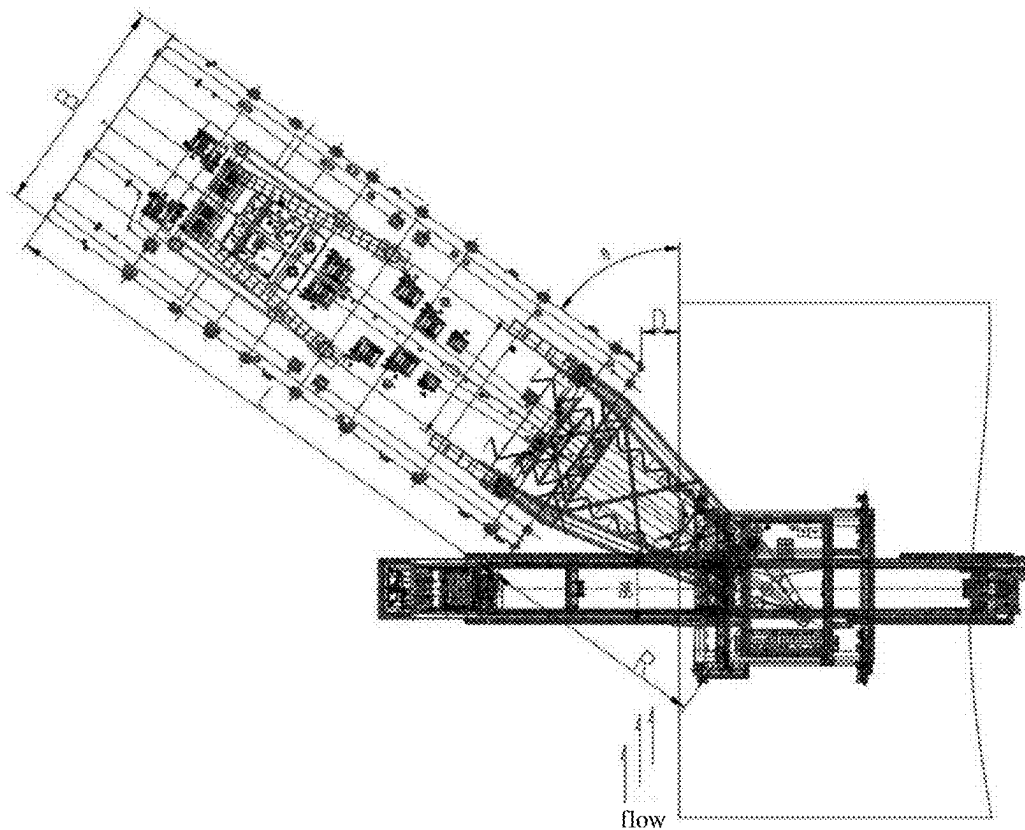


FIG. 8

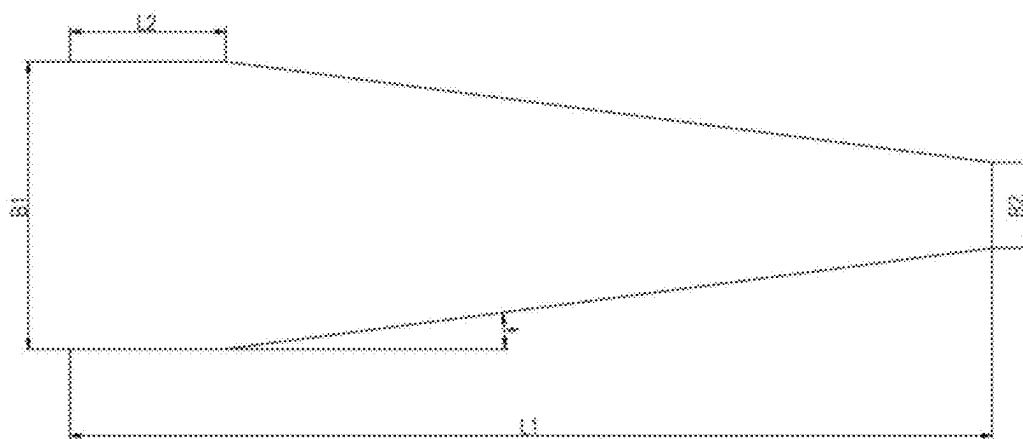
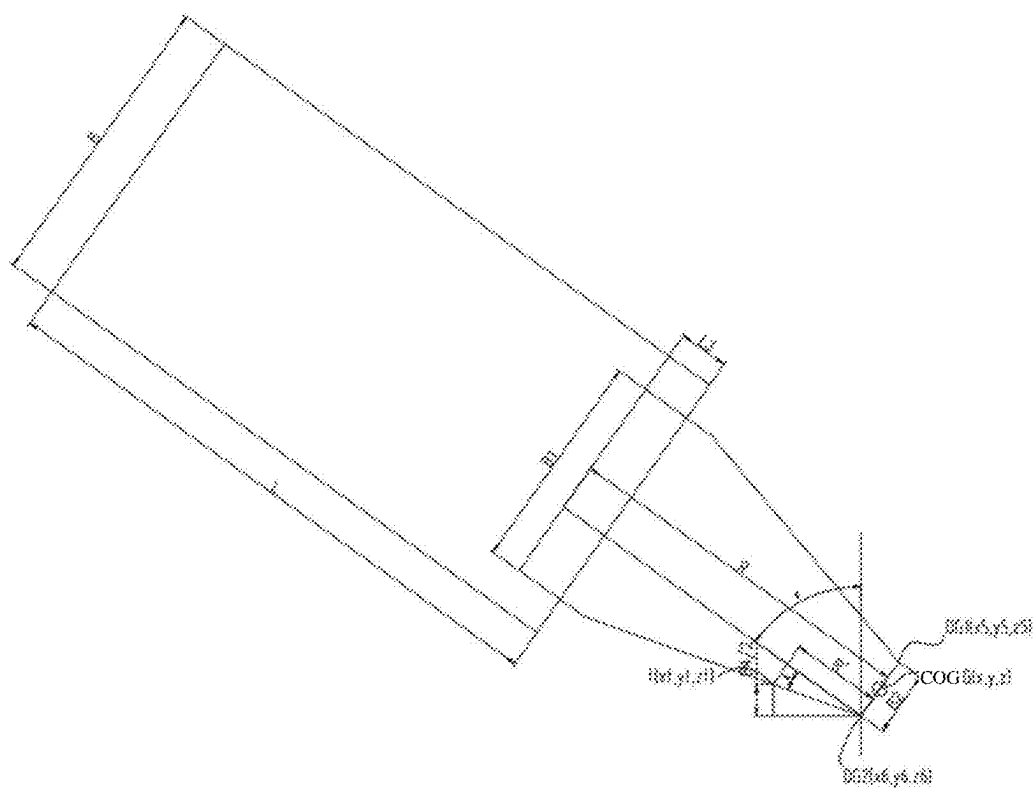


FIG. 9



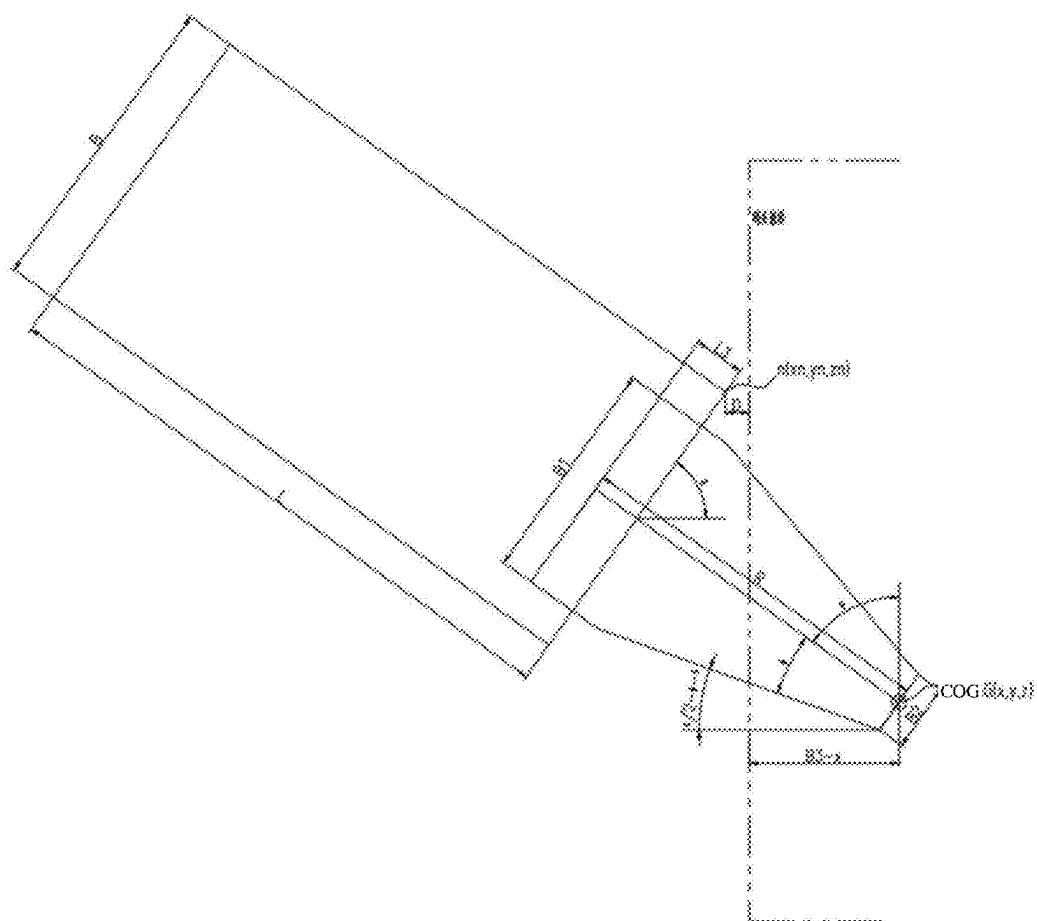


FIG. 11

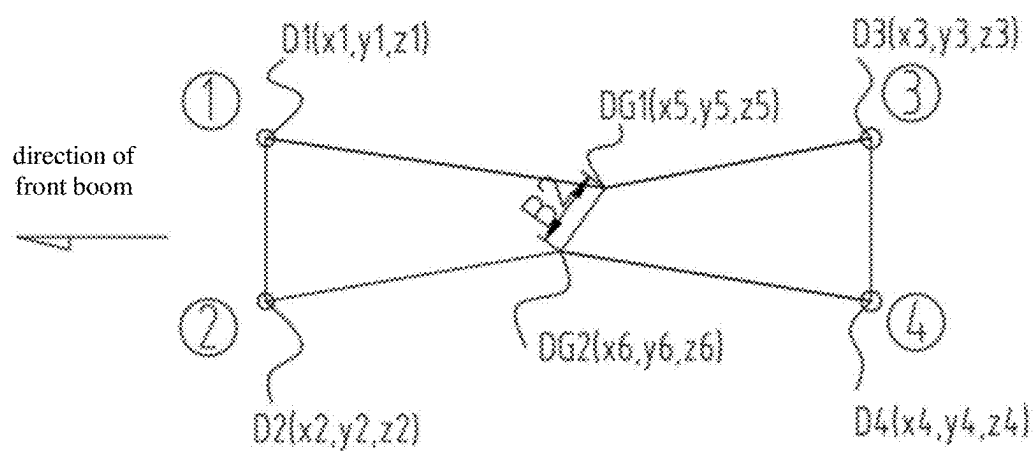


FIG. 12

FULL-UNIT LIFTING METHOD FOR QUAYSIDE CONTAINER CRANE

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] The present application claims the priority benefit of China patent application No. 202410269885.9, filed on Mar. 11, 2024. The entirety of China patent application No. 202410269885.9 is hereby incorporated by reference herein and made a part of this specification.

TECHNICAL FIELD

[0002] The present application relates to a technical field of lift of quayside container crane, and in particular, to a full-unit lifting method for a quayside container crane.

BACKGROUND

[0003] A Container Quayside Crane (abbreviated as QCC) is a primary equipment for loading and unloading containers between a container vessel and an apron. The quayside container crane primarily includes a front boom, a rear boom, a gantry frame structure, a machinery house, a front tie link, a rear tie link, a traveling mechanism and the like.

[0004] At present, there are two main methods for installing quayside container cranes after being transported to a dock.

[0005] The first method involves preliminary assembly of the quayside container crane into several major modules at a manufacturing facility, such as the front boom, the rear boom (including the machinery house, a trapezoidal frame, and a rear tie link), left and right portal legs, a lower cross beam, a traveling trolley and the like. These modules are transported to the dock and assembled on-site by using floating cranes and other lifting equipment according to a general assembly process, and key steps include: 1, marking reference lines based on the crane's gauge and base; 2, arranging lifting travelling mechanisms according to the marked lines and fixing the traveling trolley through support; 3, installing two lower cross beams, and connecting the lower cross beams to the traveling trolley through bolted joints; 4, lifting the left and right portal legs onto the lower cross beams respectively, and welding them and securing wind ropes after the left and right portal legs are vertically oriented; 5, lifting a rear boom assembly (including a rear boom, a machinery house, a trapezoidal frame, a rear tie link and an upper cross beam) onto the gantry frame and welding the rear boom assembly after horizontal orientation; 6, lifting the front boom and assembling the front boom with a hinge pin at a hinge point of the rear boom, and maintaining slight upward tilt of the front boom; 7, installing the front tie link and leveling the front boom; and 8, threading wire ropes for mechanisms, connecting junction boxes, conducting installation inspections, lubricating, and preparing for power-on debugging.

[0006] The second method involves assembling and debugging the quayside container crane full-unit at the manufacturing facility according to a general assembly process and transporting the full-unit to the user terminal. During the transportation, wheel parts of the traveling mechanism of the quayside container crane are steered by 90 degrees. The quayside container crane is winched via a

winch to the user terminal along rails laid between the terminal and transport vessel through roll-on/roll-off (RO/RO) operations.

[0007] The manufacturing facility provides superior construction conditions for full-unit assembly of the quayside container crane, whereas on-site assembly at the user terminal faces constrained construction conditions that compromise overall assembly quality. Moreover, extended durations for welding and assembly operations at the user terminal occupy valuable terminal operational areas, significantly impacting port productivity.

[0008] Although the RO/RO method reduces construction time, it imposes stringent water level requirements. This method becomes infeasible during dry seasons along the Yangtze River with a large water level differential. Additionally, RO/RO operations require extensive preparatory work, including pre-engineered wheel orientation in the traveling mechanism and pre-fabrication of specialized rail beams for the rolling process, resulting in higher costs.

[0009] While the RO/RO method remains a prevalent installation approach, its applicability is constrained by the water level of the terminal. For inland river terminals experiencing winter low-water periods (particularly with water level differentials exceeding 5 meters), implementing RO/RO operations necessitates constructing elevated rail support beams, which incurs substantial costs and complicates transportation reinforcement measures.

SUMMARY

[0010] The present application aims to provide a full-unit lifting method for a quayside container crane. After completing the installation and commissioning of the quayside container crane full-unit at the manufacturing facility, it is transported to a user terminal. During onshore transfer at the user terminal, this method eliminates dependency on terminal water levels by lateral lifting the quayside container crane full-unit via a floating crane to the terminal. Compared to the roll-on/roll-off (RO/RO) methods, this approach achieves lower costs under conditions of significant water level differentials, offers greater operational convenience, and requires substantially shorter construction time, which is typically completed within a single day.

[0011] To achieve the above object, the technical solution of the present application is as follows,

[0012] A full-unit lifting method for a quayside container crane includes following steps:

[0013] step 1: calculating an overall center position of gravity of the quayside container crane full-unit;

[0014] step 2: estimating a lifting height and a floating amplitude required by a main hook of a floating crane according to a water level differential at a dock and a height of the quayside container crane full-unit;

[0015] step 3: selecting the floating crane and obtaining detailed parameters of the floating crane according to basic lifting conditions of the floating crane;

[0016] step 4: selecting a type of the main hook of the floating crane according to a weight of the quayside container crane full-unit, checking whether an interference between the boom of the floating crane and the front boom occurs, and determining an inclination angle of the floating crane during lifting;

[0017] step 5: calculating a load on the lifting wire rope and making selection of the lifting wire rope and a shackle;

[0018] step 6: making preparations before lifting; and

[0019] step 7: performing lifting operation at the dock.

[0020] Further, the step 1 specifically includes calculating a center of gravity of each of assemblies of the quayside container crane full-unit according to a design drawing of the quayside container crane full-unit, summarizing the centers of gravity of the assemblies of the quayside container crane full-unit into a center of gravity position table, and obtaining an overall center of gravity position G (x, y, z) of the quayside container crane full-unit by summing values of the centers of gravity of the assemblies of the quayside container crane full-unit in the center of gravity position table.

[0021] Further, the center of gravity of each of the assemblies of the quayside container crane full-unit is calculated by defining an intersection of a left-right symmetry central plane of the quayside container crane full-unit, an upper plane of a rail and a longitudinal central plane of a waterside rail as an origin, and defining a direction of a front boom as an x-axis positive direction.

[0022] Further, the water level differential at the dock h_1 and the height of the quayside container crane full-unit H in the step 2 are known parameters.

[0023] Further, the basic lifting conditions in the step 3 are that a maximum lifting height of the main hook of the floating crane h_{4max} > the height of the quayside container crane full-unit H + a length of the lifting wire rope $C \times \sin \gamma$, where C is the length of the lifting wire rope and γ is an angle between the lifting wire rope and the horizontal plane, and the detailed parameters of the floating crane include a height from a deck surface of the floating crane to a water surface h_2 , a height from the deck surface of the floating crane to a lower hinge joint of a boom of the floating crane h_3 , a length of the boom of the floating crane L_1 , an angle between the boom of the floating crane and a horizontal plane α , a lifting height of the main hook of the floating crane h_4 , a height from the main hook of the floating crane to an upper pulley at the boom of the floating crane h_5 , a radius of the boom of the floating crane R, a width of the floating crane B, a length of the floating crane L, and a distance from the lower hinge joint of the boom of the floating crane to a front end of the floating crane L_3 , where $R = L_1 \times \cos \alpha$, $h_1 + h_4 + h_5 - h_2 + h_3 + L_1 \times \sin \alpha$, $h_4 + h_5$ is a constant, h_4 and h_5 are not constant values, h_5 has a minimum value h_{5min} , when $h_5 = h_{5min}$, h_4 reaches its maximum value h_{4max} .

[0024] Further, the length of each of the four lifting wire ropes C is 24 m, the angle γ between the four lifting wire ropes and the horizontal plane is 60° at minimum, and the height of the main hook to the upper pulley at the boom of the floating crane h_5 has a minimum value h_{5min} of 5 meters.

[0025] Further, the step 4 specifically includes when the weight of the quayside container crane full-unit exceeds 700 t, a floating crane with two main hooks is selected, whereby the length of the boom of the floating crane L_1 , a distance between two lower hinge joints of the floating crane B_1 , a distance between two main hooks at an uppermost end of the

floating crane B_2 , a length of a straight edge of the boom of the floating crane L_2 , and an angle ϕ between an inclined edge and the straight edge of the boom of the floating crane are satisfied with

$$\phi = \arctan \frac{B_{\text{②}} - B_2}{2(L_{\text{②}} - L_2)} \text{②}$$

② indicates text missing or illegible when filed

an angle θ between the boom of the floating crane and a dock shoreline, a center of a connecting line between the two main hooks of the floating crane coincides with the overall center of gravity position of the quayside container crane full-unit in a top view, a height of a first main hook of the two main hooks to the dock surface h_4' , and a height of a second main hook of the two main hooks to the dock surface h_4'' are satisfied with $h_4' = h_2 + h_3 + L_1 \times \sin \alpha - h_1 - h_5'$; $h_4'' = h_2 + h_3 + L_1 \times \sin \alpha - h_1 - h_5''$; a position of the first main hook is set as $DG_1 (x_5, y_5, z_5)$,

$$x_5 = x - \frac{B_2}{2} \times \sin \theta, y_5 = y + \frac{B_2}{2} \times \cos \theta, z_5 = h_4';$$

a position of the second main hook is set as $DG_2 (x_6, y_6, z_6)$,

$$x_6 = x + \frac{B_2}{2} \times \sin \theta, y_6 = y - \frac{B_2}{2} \times \cos \theta, z_6 = h_4'';$$

assuming that in the top view, an intersection between the front boom and the boom of the floating crane is $t(x_t, y_t, z_t)$, it is obtained as:

$$y_{\text{②}} - y_{\text{③}} = \frac{B_2}{2}, y_{\text{③}} = y_6 - \frac{B_4}{2}, \frac{y_6 - y_1}{x_{\text{②}} - x_6} = \tan\left(\frac{\pi}{2} - \theta - \phi\right),$$

$$x_{\text{②}} = x_6 + \frac{B_4}{2 \tan\left(\frac{\pi}{2} - \theta - \phi\right)},$$

$$R' = \frac{y_{\text{②}} - y_{\text{③}}}{\sin\left(\frac{\pi}{2} - \theta - \phi\right)} \cos \phi = \frac{B_4}{2 \sin\left(\frac{\pi}{2} - \theta - \phi\right)} \cos \phi, z_t = z_6 + h_5 \text{②} R' \times \cot \alpha,$$

② indicates text missing or illegible when filed

whereby a basic condition for judging whether the interference between the boom of the floating crane and the front boom occurs is $z_t >$ the height of the front boom of the quayside container crane h_6 , when the interference occurs, a re-check is performed by adjusting the angle θ between the boom of the floating crane and the dock, a distance D

between the floating crane and an apron is checked, a corner point of the floating crane closest to the apron is set as $n(x_n, y_n, z_n)$, D is obtained by calculating x_n ,

$$x_{\textcircled{2}} = x + (R - L_{\textcircled{2}}) \times \cos\left(\frac{\pi}{2} - \theta - \phi\right) - \frac{B}{2} \times \cos\theta,$$

$$D = x_{\textcircled{2}} - B_{\textcircled{2}} = x + (R - L_{\textcircled{2}}) \times \cos\left(\frac{\pi}{2} - \theta - \phi\right) - \frac{B}{2} \times \cos\theta - B_{\textcircled{2}},$$

$\textcircled{2}$ indicates text missing or illegible when filed

in order to prevent the floating crane from colliding due to a fact that the floating crane is too close to the apron during lifting, D possesses a minimum safe distance, and a basic condition for determining whether a safe distance between the floating crane and the apron is enough is $D > 2$ m.

[0026] Further, the step 5 specifically includes the position of the first main hook is $DG_1(x_5, y_5, z_5)$, and

$$x_5 = x - \frac{B_2}{2} \times \sin\theta, y_5 = y + \frac{B_2}{2} \times \cos\theta, z_5 = h_4',$$

the position of the second main hook is $DG_2(x_6, y_6, z_6)$, and

$$x_6 = x + \frac{B_2}{2} \times \sin\theta, y_6 = y - \frac{B_2}{2} \times \cos\theta, z_6 = h_4'',$$

positions of four lifting lugs at the quayside container crane full-unit are obtained by measuring dimensions on the design drawing as: $D_1(x_1, y_1, z_1)$, $D_2(x_2, y_2, z_2)$, $D_3(x_3, y_3, z_3)$, $D_4(x_4, y_4, z_4)$, lengths of four lifting wire ropes are respectively calculated as follows: the length of a first lifting wire rope of the four lifting wire ropes is $C_1 = \sqrt{(x_5 - x_1)^2 + (y_5 - y_1)^2 + (z_5 - z_1)^2}$, the length of a second lifting wire rope of the four lifting wire ropes is $C_2 = \sqrt{(x_6 - x_2)^2 + (y_6 - y_2)^2 + (z_6 - z_2)^2}$, the length of a third lifting wire rope of the four lifting wire ropes is $C_3 = \sqrt{(x_5 - x_3)^2 + (y_5 - y_3)^2 + (z_5 - z_3)^2}$, the length of a fourth lifting wire rope of the four lifting wire ropes is $C_4 = \sqrt{(x_6 - x_4)^2 + (y_6 - y_4)^2 + (z_6 - z_4)^2}$; the height h_4' of the first main hook and the height h_4'' of the second main hook are calculated by setting the length C_2 of the second lifting wire rope and the length C_3 of the third lifting wire rope, so as to obtain the length C_1 of the first lifting wire rope and the length C_4 of the fourth lifting wire rope, and then the angle γ between each of the four lifting wire ropes and a horizontal plane exceeds 60° , the angle between each of the four lifting wire ropes and the horizontal plane is calculated as follows:

$$\gamma_{\textcircled{2}} = \arcsin\left(\frac{z_5 - z_{\textcircled{2}}}{C_1}\right), \gamma_{\textcircled{2}} = \arcsin\left(\frac{z_6 - z_{\textcircled{2}}}{C_2}\right),$$

$$\gamma_3 = \arcsin\left(\frac{z_5 - z_3}{C_3}\right), \gamma_4 = \arcsin\left(\frac{z_6 - z_4}{C_4}\right) \textcircled{2}$$

$\textcircled{2}$ indicates text missing or illegible when filed

and then the load on each of the four lifting wire rope is calculated as follows: since the gravity of the floating crane

is vertically downward and the center of the connecting line between the two main hooks coincides with the overall center of gravity position of the quayside container crane full-unit in the top view, a load on the first main hook and a load on the second main hook is substantially balanced, it is obtained that a vertical downward load on each of the two main hooks is

$$\frac{Q}{2}$$

when a total weight Q of the quayside container crane is obtained according to the design drawing, so that the load on the first lifting wire rope, the load on the second lifting wire rope, the load on the third lifting wire rope and the load on the fourth lifting wire rope are respectively calculated with the first lifting wire rope C_1 and the third lifting wire rope C_3 sharing a first main hook while the second lifting wire rope C_2 and the fourth lifting wire rope C_4 sharing the second main hook, based on that a resultant force in a vertical direction of two of the four lifting wire ropes of one of the two main hooks is a lifting weight

$$\frac{Q}{2},$$

and a resultant force in a horizontal direction is 0, it is obtained that

$$F_{\textcircled{2}} \times \sin\gamma_{\textcircled{2}} + F_{\textcircled{2}} \times \sin\gamma_{\textcircled{2}} = \frac{Q}{2},$$

$\textcircled{2}$ indicates text missing or illegible when filed

$$F_1 \times \cos\gamma_1 = F_3 \times \cos\gamma_3;$$

$$F_{\textcircled{2}} \times \sin\gamma_{\textcircled{2}} + F_4 \times \sin\gamma_4 = \frac{Q}{2}, F_2 \times \cos\gamma_2 = F_4 \times \cos\gamma_4,$$

$\textcircled{2}$ indicates text missing or illegible when filed

the load on the first lifting wire rope, the load on the second lifting wire rope, the load on the third lifting wire rope and the load on the fourth lifting wire rope are obtained by solving equations respectively:

$$F_{\textcircled{2}} = \frac{Q}{2(\sin\gamma_{\textcircled{2}} + \cos\gamma_{\textcircled{2}} \times \tan\gamma_{\textcircled{2}})}, F_{\textcircled{2}} = \frac{Q}{2(\sin\gamma_{\textcircled{2}} + \cos\gamma_{\textcircled{2}} \times \tan\gamma_{\textcircled{2}})},$$

$$F_{\textcircled{2}} = \frac{Q}{2(\sin\gamma_{\textcircled{2}} + \cos\gamma_{\textcircled{2}} \times \tan\gamma_{\textcircled{2}})}, F_{\textcircled{2}} = \frac{Q}{2(\sin\gamma_{\textcircled{2}} + \cos\gamma_{\textcircled{2}} \times \tan\gamma_{\textcircled{2}})}$$

$\textcircled{2}$ indicates text missing or illegible when filed

according to the load on the first lifting wire rope, the load on the second lifting wire rope, the load on the third lifting wire rope and the load on the fourth lifting wire rope, a lifting wire rope having a diameter with a safety factor above

4 and a shackle with a safety coefficient above 1 are selected by referencing a wire rope breaking force table.

[0027] Further, the step 6 specifically includes before lifting, travelling a trolley to a maximum rear extension position of the rear boom, retracting an upper frame of a container hanger to its highest position, parking an elevator at a first floor, locking brakes of the assemblies in such a way, that the trolley and trailer trolley wheels are plugged by wedge blocks, to keep the front boom be horizontal; for a travelling mechanism, inserting wooden blocks at rotatable positions of eight-wheel equalizer beams and driving and driven bogies before the lifting, so as to prevent rotation; checking securing and fastening of the assemblies of the quayside container crane to ensure all movable components are fastened; disassembling all anchorages between the quayside container crane and a transport vessel; using two winches on a floating crane vessel to secure two portal legs obliquely opposite the quayside container crane before the lifting, diagonal portal legs of the quayside container crane are pulled by the two winches on the floating crane vessel after the lifting, to maintain a constant inclination angle between the floating crane and the quayside container crane.

[0028] Further, the step 7 of performing lifting operation at the dock further includes: 1) after a floating crane vessel arrives at a construction site, the floating crane vessel is positioned within 200 m offshore from a quay front line, the floating crane vessel inclines, a length of an anchor cable is adjusted with reference to barge parameters and water conditions of the dock, to realize operational processes of lifting, shifting, and parking; 2) after the floating crane vessel is positioned, all tools and equipment used during the lifting are comprehensively checked, the floating crane vessel is put into use after it is confirmed that no abnormal condition exists, and whether a lifting point of the equipment is consistent with a diameter of a shackle pin or not is checked, after checking, the angle of the boom is adjusted to a required angle, all mechanical parts of the floating crane vessel are rechecked, and the floating crane vessel is used after being confirmed to be intact; 3) the floating crane vessel is lifted by the two main hooks, the two main hooks are loosened above the equipment to be lifted, rigging suspended on the two main hooks in advance is attached to lifting lugs of the equipment in sequence, and reliability of each connection point is rechecked; 4) after preparation, the lifting is performed, when the lifting is 200 mm away, all starting equipment brakes, a component is suspended on the two main hooks for secondary braking, a static state is maintained, when there is no abnormal condition, the floating crane is started to lift the component to a preset height and is suspended in a static state, the object is horizontally lifted during the lifting, to avoid uneven height discrepancies between the two main hooks; 5) after the component is lifted, a towing belt of a tugboat is used to assist in shifting to an installation berth, and anchor positioning is performed at an installation site, the floating crane vessel is repositioned after arriving at the installation site by adjusting a length and an orientation of the anchor cable, so that the floating crane is positioned above the rail of the installation dock, and the floating crane is kept descending horizontally, so that the equipment in a suspended state is aligned to a position right above a placing point by 10 cm; 6) a placing position of the equipment is approved, the floating crane is commanded to loosen the two main hooks, and the equipment is placed at the placing point; 7) lifting construction is

carried out until the lifting is finished, and all lifting tools are retracted and the floating crane vessel is withdrawn.

[0029] Compared with the existing technique, the present application can achieve at least one of the following beneficial effects:

[0030] 1. According to the full-unit lifting method for the quayside container crane provided by the present application, the onshore installation time of the quayside container crane full-unit at the dock is significantly reduced. The quayside container crane full-unit may be lifted onto the dock within a single day, which minimizes dock occupancy and avoids disruptions to production.

[0031] 2. With the angle between the floating crane and the front boom between 40 degrees to 50 degrees, center-of-gravity deviations of the quayside container crane are dynamically compensated by adjusting the position of the trolley along the boom. This enables reusable deployment of lifting accessories (e.g., wire ropes and shackles), making the method universally applicable to lifting operations of cranes of the same type.

[0032] 3. The full-unit lifting method for the quayside container crane provided by the present application demonstrates high reliability, strong operability, and broad applicability. It can be directly extended to crane relocation on the dock and vessel unloader lifting operations with identical operational principles.

BRIEF DESCRIPTION OF THE DRAWINGS

[0033] FIG. 1 is a schematic diagram of a structure of a quayside container crane full-unit according to the present application;

[0034] FIG. 2 is a schematic diagram of structural parts of the quayside container crane full-unit according to the present application;

[0035] FIG. 3 is a schematic diagram of an RO-on/RO-off process for the quayside container crane full-unit according to the present application;

[0036] FIG. 4 is a flow chart of a full-unit lifting method for the quayside container crane according to the present application;

[0037] FIG. 5 is a gravity center position table of the quayside container crane full-unit according to the present application;

[0038] FIG. 6 shows a gravity center position and a coordinate system of the quayside container crane full-unit according to the present application;

[0039] FIG. 7 schematically shows a first relevant parameter of lifting by a floating crane according to the present application;

[0040] FIG. 8 schematically shows a second relevant parameter of the lifting by the floating crane according to the present application;

[0041] FIG. 9 shows relevant parameters of a boom of the floating crane according to the present application;

[0042] FIG. 10 shows relevant parameters for determining whether the floating crane interferes with the front boom according to the present application;

[0043] FIG. 11 shows relevant parameters for determining whether there is a sufficient safe distance between the floating crane and an apron according to the present application; and

[0044] FIG. 12 is a top view of a lifting wire rope according to the present application.

DETAILED DESCRIPTION

[0045] The present application will be further described in detail below in combination with the accompanying drawings and embodiments.

[0046] Referring to FIGS. 1 to 12, a full-unit lifting method for a quayside container crane specifically includes following steps.

[0047] 1. Calculating an overall center of gravity (COG) of the quayside container crane: calculating a COG of each assembly of the quayside container crane full-unit according to a design drawing of the quayside container crane full-unit, by which an origin is defined as an intersection of a left-right symmetry central plane of the quayside container crane full-unit, an upper plane of a rail and a longitudinal central plane of a waterside rail, a direction of a front boom is defined an x-axis positive direction, summarizing the COG of the assemblies of the quayside container crane full-unit into a COG table, and obtaining the overall COG G (x, y, z) of the quayside container crane full-unit by summing values of the COG of the assemblies of the quayside container crane full-unit in the COG table.

[0048] 2. Estimating a lifting height and a floating amplitude required by a hook of a floating crane according to a water level differential at a dock and a height of the quayside container crane full-unit, by which the floating crane is selected and detailed parameters thereof are obtained according to basic lifting conditions of the floating crane, and an outline diagram of the floating crane is drawn; known parameters include the water level differential at the dock h_1 , the overall COG position of the quayside container crane full-unit G (x, y, z), position coordinates of four lifting lugs on the quayside container crane D1 (x_1, y_1, z_1), D2 (x_2, y_2, z_2), D3 (x_3, y_3, z_3), D4 (x_4, y_4, z_4), a height of the quayside container crane full-unit H, a height from the front boom to the plane of the rail h_6 , a distance from the waterside rail to the apron B_3 , a width of the front boom B_4 ; relevant parameters of the floating crane include a height from a deck surface of the floating crane to a water surface h_2 , a height from the deck surface of the floating crane to a lower hinge joint of a boom of the floating crane h_3 , a length of the boom of the floating crane L_1 , an angle between the boom of the floating crane and a horizontal plane α , a lifting height of the hook of the floating crane h_4 , a height from the hook of the floating crane to an upper pulley at the boom of the floating crane h_5 , a radius of the boom of the floating crane R, a width of the floating crane B, a length of the floating crane L, and a distance from the lower hinge joint of the boom of the floating crane to a front end of the floating crane L_3 , where $R=L_1 \times \cos \alpha$, generally α is 65° ; $h_1+h_4+h_5=h_2+h_3+L_1 \times \sin \alpha$, h_4+h_5 is a constant. However, since the hook may move up and down through a lifting mechanism, h_4 and h_5 are not constant values, h_5 has a minimum value h_{5min} which is generally set to 5 m due to a minimum safe distance between the hook and the upper pulley of the boom; when $h_5=h_{5min}$, h_4 reaches its maximum value h_{4max} , which is a basic condition for determining whether the floating crane may perform the lifting, namely a maximum lifting height of the hook of the floating crane h_{4max} > the height of the quayside container crane full-unit H + a length of the lifting wire rope $C \times \sin \gamma$, where C is the length of the lifting wire rope and γ is an angle between the lifting wire rope and the horizontal plane, generally, the length of the lifting wire rope C is taken as 24 m, and the angle γ between the lifting wire rope and the horizontal plane is 60° at minimum.

[0049] 3. Checking interference between the boom of the floating crane and the front boom, and determining an inclination angle of the floating crane during lifting. Since a weight of the quayside container crane full-unit typically exceeds 700 tons, the floating crane with two main hooks is required. The relevant parameters of the floating crane include the length of the boom of the floating crane L_1 , a distance between two lower hinge joints of the floating crane B_1 , a distance between two main hooks at an uppermost end of the floating crane B_2 , a length of a straight edge of the boom of the floating crane L_2 , and an angle ϕ between an inclined edge and the straight edge of the boom of the floating crane with

$$\phi = \arctan \frac{B_1 - B_2}{2(L_1 - L_3)};$$

as the two main hooks of the floating crane are adopted for lifting, the angle between the boom of the floating crane and the dock shoreline is θ , a center of a connecting line between the two hooks of the floating crane must coincide with the overall COG position of the quayside container crane full-unit in the top view. The lifting heights of the two hooks may be different (with a height difference not too large). It is assumed that a height from a first hook of the floating crane to the dock surface is h_4' , and a height from a second hook of the floating crane to the dock surface is h_4'' , $h_4' = h_2 + h_3 + L_1 \times \sin \alpha - h_1 - h_5'$; $h_4'' = h_2 + h_3 + L_1 \times \sin \alpha - h_1 - h_5''$. If a position of the first hook of the floating crane is set as DG₁ (x_5, y_5, z_5),

$$x_5 = x - \frac{B_2}{2} \times \sin \theta, y_5 = y + \frac{B_2}{2} \times \cos \theta, z_5 = h_4';$$

if a position of the second hook of the floating crane is set as DG₂ (x_6, y_6, z_6),

$$x_6 = x + \frac{B_2}{2} \times \sin \theta, y_6 = y - \frac{B_2}{2} \times \cos \theta, z_6 = h_4'';$$

assuming that in the top view, an intersection between the front boom and the boom of the floating crane is t(x_t, y_t, z_t), the following may be obtained as:

$$\begin{aligned} y_6 - y_t &= \frac{B_4}{2}, \\ y_t &= y_6 - \frac{B_4}{2}, \\ \frac{y_6 - y_t}{x_t - x_6} &= \tan\left(\frac{\pi}{2} - \theta - \phi\right), \\ x_t &= x_6 + \frac{B_4}{2 \tan\left(\frac{\pi}{2} - \theta - \phi\right)}, \\ R' &= \frac{y_6 - y_t}{\sin\left(\frac{\pi}{2} - \theta - \phi\right)} \cos \phi = \frac{B_4}{2 \sin\left(\frac{\pi}{2} - \theta - \phi\right)} \cos \phi, \\ z_t &= z_6 + h_5 - R' \times \cot \alpha, \end{aligned}$$

and the basic condition for determining the interference between the boom of the floating crane and the front boom includes: $z_r >$ the height of the front boom of the quayside container crane h_6 ; if interference occurs, re-check is performed by adjusting the angle θ between boom of the floating crane and the dock, where θ cannot be adjusted arbitrarily; θ is ensured not too small to avoid collision between a floating crane vessel and the dock, and thus it is also necessary to check a distance D between the floating crane and the apron; it is assumed that a corner point of the floating crane closest to the apron is $n(x_n, y_n, z_n)$, and D can be obtained by calculating x_n ;

$$x_n = x + (R - L_3) \times \cos\left(\frac{\pi}{2} - \theta - \phi\right) - \frac{B}{2} \times \cos\theta,$$

$$D = x_n - B_3 = x + (R - L_3) \times \cos\left(\frac{\pi}{2} - \theta - \phi\right) - \frac{B}{2} \times \cos\theta - B_3,$$

in order to prevent the floating crane from colliding due to the fact that the floating crane is too close to the apron during lifting, D possesses the minimum safety distance, and the basic condition for determining whether a safe distance between the floating crane and the apron is enough is $D > 2$ m.

[0050] 4. Calculating a load on the wire rope and selecting the wire rope and shackles; the position of the first hook of the floating crane is $DG_1(x_5, y_5, z_5)$ and

$$x_5 = x - \frac{B_2}{2} \times \sin\theta, y_5 = y + \frac{B_2}{2} \times \cos\theta, z_5 = h_4',$$

the position of the second hook of the floating crane is $DG_2(x_6, y_6, z_6)$, and

$$x_6 = x + \frac{B_2}{2} \times \sin\theta, y_6 = y - \frac{B_2}{2} \times \cos\theta, z_6 = h_4'',$$

the positions of four lifting lugs at the quayside container crane are obtained by measuring dimensions on the drawings as: $D_1(x_1, y_1, z_2)$, $D_2(x_2, y_2, z_2)$, $D_3(x_3, y_3, z_3)$, $D_4(x_4, y_4, z_4)$; then lengths of the four lifting wire ropes are as follows:

[0051] the length of a first lifting wire rope is $C_1 = \sqrt{(x_5 - x_1)^2 + (y_5 - y_1)^2 + (z_5 - z_1)^2}$, the length of a second lifting wire rope is $C_2 = \sqrt{(x_6 - x_2)^2 + (y_6 - y_2)^2 + (z_6 - z_2)^2}$, the length of a third lifting wire rope is $C_3 = \sqrt{(x_5 - x_3)^2 + (y_5 - y_3)^2 + (z_5 - z_3)^2}$, the length of a fourth lifting wire rope is $C_4 = \sqrt{(x_6 - x_4)^2 + (y_6 - y_4)^2 + (z_6 - z_4)^2}$. The heights h_4' and h_4'' of the two hooks of the floating crane can be calculated by setting the length C_2 of the second lifting wire rope and the length C_3 of the third lifting wire rope, and the length C_1 of the first lifting wire rope and the length C_4 of the fourth lifting wire rope can be obtained; checking that the angle γ between

each of the lifting wire ropes and the horizontal plane exceeds 60° . The angle between each of the lifting wire ropes and the horizontal plane is as follows:

$$\gamma_1 = \arcsin\left(\frac{z_5 - z_1}{C_1}\right), \gamma_2 = \arcsin\left(\frac{z_6 - z_2}{C_2}\right),$$

$$\gamma_3 = \arcsin\left(\frac{z_5 - z_3}{C_3}\right), \gamma_4 = \arcsin\left(\frac{z_6 - z_4}{C_4}\right);$$

The load on the lifting wire rope is calculated as follows: since the gravity is vertically downward, and since the center of the two hooks of the floating crane corresponds to the overall COG position of the quayside container crane full-unit, it can be considered that the loads on the two hooks of the floating crane is substantially balanced. A total weight Q of the quayside container crane may be known according to the design drawing, and it may be obtained that a vertical downward load on each of the hooks of the floating crane may be

$$\frac{Q}{2},$$

so that the load on each of the lifting wire ropes may be calculated as follows. The first lifting wire rope C_1 and the third lifting wire rope C_3 share one hook of the floating crane, and the second lifting wire rope C_2 and the fourth lifting wire rope C_4 share the other hook of the floating crane. According to that a resultant force in the vertical direction of the two lifting wire ropes of the same hook of the floating crane is a lifting weight

$$\frac{Q}{2},$$

and a resultant force in the horizontal direction should be 0, it may be obtained that

$$F_1 \times \sin\gamma_1 + F_3 \times \sin\gamma_3 = \frac{Q}{2},$$

$$F_1 \times \cos\gamma_1 = F_3 \times \cos\gamma_3;$$

$$F_2 \times \sin\gamma_2 + F_4 \times \sin\gamma_4 = \frac{Q}{2},$$

$F_2 \times \cos\gamma_2 = F_4 \times \cos\gamma_4$. The loads on the four wire ropes may be obtained by solving equations respectively

$$F_1 \frac{Q}{2(\sin\gamma_1 + \cos\gamma_1 \times \tan\gamma_3)}, F_3 \frac{Q}{2(\sin\gamma_3 + \cos\gamma_3 \times \tan\gamma_1)},$$

$$F_2 \frac{Q}{2(\sin\gamma_2 + \cos\gamma_2 \times \tan\gamma_4)}, F_4 \frac{Q}{2(\sin\gamma_4 + \cos\gamma_4 \times \tan\gamma_2)}$$

[0052] According to the loads on the wire ropes, a wire rope having a diameter with a safety factor above 4 and a

shackle with a safety coefficient above 1 are selected by referencing a wire rope breaking force table.

[0053] Specifications and lengths of the wire rope and types of the shackle and the like are compiled into a table and are in one-to-one correspondence with the lifting lugs.

[0054] Example: length parameters of the wire rope are shown in the following table:

position	length	diameter	theoretical load	type of shackle
1	22.6 m	150 mm	280 t	300 t
2	22 m	150 mm	330 t	350 t
3	22 m	150 mm	337 t	350 t
4	24 m	150 mm	277 t	300 t

[0055] 5. Making preparations before lifting: before the lifting, travelling a trolley to a maximum rear extension position of the rear boom (adjustable based on the specific COG position); retracting an upper frame of a container hanger to its highest position, parking an elevator at a first floor, locking brakes of respective mechanisms, plugging the trolley and trailer trolley wheels by wedge blocks, and keeping the front boom horizontal; for a travelling mechanism, inserting wooden blocks at rotatable positions of the eight-wheel equalizer beams and driving and driven bogies before the lifting, so as to prevent rotation; checking the securing and fastening of all components of the quayside container crane to ensure all movable components are fastened firmly; disassembling all anchorages between the quayside container crane and the transport vessel; using two winches on the floating crane vessel to secure two portal legs obliquely opposite the quayside container crane before lifting (through the pre-adjusted lengths of the wire ropes, rotation due to insufficient horizontal stability after the quayside container crane may be prevented after the quayside container crane is hoisted, and the diagonal legs are pulled by the winches for protection), the diagonal portal legs of the quayside container crane must be pulled by the two winches on the floating crane vessel continuously after lifting, to maintain a constant inclination angle between the floating crane and the quayside container crane.

[0056] 6. Performing lifting operation at the user dock including following steps.

[0057] 1) After the floating crane arrives at a construction site, the floating crane is positioned within about 200 m offshore from the quay front line, the floating crane inclines at a certain angle (generally, an angle between an extension direction of the boom of the floating crane and the dock is) 50°, a length of an anchor cable is adjusted with reference to barge parameters and the water conditions of the dock, to ensure that operational processes of lifting, shifting, parking and the like may be achieved.

[0058] 2) After the floating crane vessel is positioned, all tools and equipment used during lifting are comprehensively checked, the floating crane vessel may be put into use after it is confirmed that no abnormal condition exists, and whether a lifting point of the equipment is consistent with a diameter of a shackle pin or not is checked. After all of the checks are performed, the angle of the boom is adjusted to a required angle. All mechanical parts of the floating crane are rechecked, and the floating crane may be used after being confirmed to be intact.

[0059] 3) The floating crane vessel are lifted by two main hooks, the hooks are loosened above equipment to be

hoisted, rigging suspended on the hooks in advance is attached to lifting lugs of the equipment in sequence, and the reliability of each connection point is rechecked.

[0060] 4) After it is confirmed that the preparation work is completed, the lifting operation is performed. When the lifting is 200 mm away, all starting equipment brakes, a component is suspended on the main hooks for secondary braking, and a static state is maintained. When there is no abnormal condition, the crane is started to hoist the component to a preset height and is suspended in the static state. The object is kept horizontally hoisted and slowly lifted in the lifting process, avoiding uneven height discrepancies between the two hooks.

[0061] 5) After the component is stably hoisted, a towing belt of a tugboat is used to assist in slow shifting to the installation berth, and the component is anchored and positioned on the installation site. The floating crane vessel is repositioned after arriving at the installation site by adjusting the length and the orientation of the anchor cable. The crane is positioned above the rail of the installation dock, and kept descending horizontally and slowly, so that the equipment in the suspended state is aligned to the position right above the placing point by 10 cm.

[0062] 6) the placing position of the equipment is approved, the floating crane is commanded to slowly loosen the hooks, and the equipment is stably placed at the placing point.

[0063] 7) According to the construction process, lifting construction is carried out until the lifting is finished, and all lifting tools are retracted and the floating crane vessel is withdrawn.

[0064] The above description includes only preferred embodiments of the present application, and is not intended to limit the present application in any form, and therefore, any modifications, equivalent substitutions and improvements made to the above embodiments according to the technical principle of the present application, without departing from the scope of the technical solution of the present application, still fall within the protection scope of the technical solution of the present application.

What is claimed is:

1. A full-unit lifting method for a quayside container crane, comprising following steps:

step 1: calculating an overall center of gravity of the quayside container crane, comprising calculating a center of gravity of each of assemblies of the quayside container crane according to a design drawing of the quayside container crane, summarizing the centers of gravity of the assemblies of the quayside container crane into a center of gravity position table, and obtaining an overall center of gravity position ($G(x, y, z)$) of the quayside container crane by summing values of the centers of gravity of the assemblies of the quayside container crane in the center of gravity position table, wherein the center of gravity of each of the assemblies of the quayside container crane is calculated by defining an intersection of a left-right symmetry central plane of the quayside container crane, an upper plane of a rail and a longitudinal central plane of a waterside rail as an origin, and defining a direction of a front boom as an x-axis positive direction;

step 2: estimating a lifting height and a floating amplitude required by a main hook of a floating crane according to a water level differential at a dock and a height of the

quayside container crane, wherein the water level differential at the dock (h_1) and the height of the quayside container crane (H) are known parameters;

step 3: selecting the floating crane and obtaining parameters of the floating crane according to basic lifting conditions of the floating crane, wherein the basic lifting conditions are that a maximum lifting height of the main hook of the floating crane (h_{4max}) is greater than the height of the quayside container crane (H)+ $C \times \sin \gamma$, where C is a length of a lifting wire rope and γ is an angle between the lifting wire rope and a horizontal plane, and the parameters of the floating crane comprise a height from a deck surface of the floating crane to a water surface (h_2), a height from the deck surface of the floating crane to a lower hinge joint of a boom of the floating crane (h_3), a length of the boom of the floating crane (L_1), an angle between the boom of the floating crane and the horizontal plane (α), the lifting height of the main hook of the floating crane (h_4), a height from the main hook of the floating crane to an upper pulley at the boom of the floating crane (h_5), a radius of the boom of the floating crane (R), a width of the floating crane (B), a length of the floating crane (L), and a distance from the lower hinge joint of the boom of the floating crane to a front end of the floating crane (L_3), where $R=L_1 \times \cos \alpha$, $h_1+h_4+h_5=h_2+h_3+L_1 \times \sin \alpha$, h_4+h_5 is a constant, h_4 and h_5 are not constant values, h_5 has a minimum value h_{5min} , when $h_5=h_{5min}$, h_4 reaches its maximum value h_{4max} .

step 4: selecting a type of the main hook of the floating crane according to a weight of the quayside container crane, checking whether an interference between the boom of the floating crane and the front boom occurs, and determining an inclination angle of the floating crane during lifting, comprising: when the weight of the quayside container crane exceeds 700 t, a floating crane with two main hooks is selected, wherein the length of the boom of the floating crane (L_1), a distance between two lower hinge joints of the floating crane (B_1), a distance between the two main hooks at an uppermost end of the floating crane (B_2), a length of a straight edge of the boom of the floating crane (L_2), and an angle (ϕ) between an inclined edge and the straight edge of the boom of the floating crane are satisfied with

$$\phi = \arccos \frac{B_1 - B_2}{2(L_1 - L_3)};$$

an angle between the boom of the floating crane and a dock shoreline (θ), a center of a connecting line between the two main hooks of the floating crane coincides with the overall center of gravity position of the quayside container crane in a top view, a height of a first main hook of the two main hooks to the dock surface (h_4'), and a height of a second main hook of the two main hooks to the dock surface (h_4'') are satisfied with $h_4' = h_2 + h_3 + L_1 \times \sin \alpha - h_1 - h_5'$; $h_4'' = h_2 + h_3 + L_1 \times \sin \alpha - h_1 - h_5''$; a position of the first main hook is set as $DG_1 (x_5, y_5, z_5)$,

$$x_5 = x - \frac{B_2}{2} \times \sin \theta, y_5 = y + \frac{B_2}{2} \times \cos \theta, z_5 = h_4';$$

a position of the second main hook is set as $DG_2 (x_6, y_6, z_6)$,

$$x_6 = x + \frac{B_2}{2} \times \sin \theta, y_6 = y - \frac{B_2}{2} \times \cos \theta, z_6 = h_4'';$$

assuming that in the top view, an intersection between the front boom and the boom of the floating crane is $t(x_t, y_t, z_t)$, it is obtained as:

$$y_{\textcircled{2}} - y_{\textcircled{2}} = \frac{B_4}{2}, y_{\textcircled{2}} = y_{\textcircled{2}} - \frac{B_4}{2}, \frac{y_{\textcircled{2}} - y_{\textcircled{2}}}{x_{\textcircled{2}} - x_{\textcircled{2}}} = \tan\left(\frac{\pi}{2} - \theta - \phi\right),$$

$$x_{\textcircled{2}} = x_{\textcircled{2}} + \frac{B_{\textcircled{2}}}{2 \tan\left(\frac{\pi}{2} - \theta - \phi\right)},$$

$$R' = \frac{y_{\textcircled{2}} - y_{\textcircled{2}}}{\sin\left(\frac{\pi}{2} - \theta - \phi\right)} \cos \phi = \frac{B_{\textcircled{2}}}{2 \sin\left(\frac{\pi}{2} - \theta - \phi\right)} \cos \phi, z_t = z_6 + h_{5\textcircled{2}} R' \cot \alpha,$$

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wherein a basic condition for judging whether the interference between the boom of the floating crane and the front boom occurs is z_t is greater than the height of the front boom of the quayside container crane (h_6), when the interference occurs, a re-check is performed by adjusting the angle (θ) between the boom of the floating crane and the dock, a distance (D) between the floating crane and an apron is checked, a corner point of the floating crane closest to the apron is set as n (x_n, y_n, z_n), D is obtained by calculating x_n ,

$$x_{\textcircled{2}} = x + (R - L_{\textcircled{2}}) \times \cos\left(\frac{\pi}{2} - \theta - \phi\right) - \frac{B}{2} \times \cos \theta,$$

$$D = x_{\textcircled{2}} - B_{\textcircled{2}} = x + (R - L_{\textcircled{2}}) \times \cos\left(\frac{\pi}{2} - \theta - \phi\right) - \frac{B}{2} \times \cos \theta - B_{\textcircled{2}}.$$

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in order to prevent the floating crane from colliding due to a fact that the floating crane is too close to the apron during lifting, D possesses a minimum safe distance, and a basic condition for determining whether a safe distance between the floating crane and the apron is enough is $D > 2$ m;

step 5: calculating a load on the lifting wire rope and making selection of the lifting wire rope and a shackle, the position of the first main hook is $DG_1 (x_5, y_5, z_5)$, and

$$x_5 = x - \frac{B_2}{2} \times \sin \theta, y_5 = y + \frac{B_2}{2} \times \cos \theta, z_5 = h_4';$$

the position of the second main hook is $DG_2 (x_6, y_6, z_6)$, and

$$x_6 = x + \frac{B_2}{2} \times \sin \theta, y_6 = y - \frac{B_2}{2} \times \cos \theta, z_6 = h_4'';$$

positions of four lifting lugs at the quayside container crane are obtained by measuring dimensions on the design drawing as: $D_1 (x_1, y_1, z_1)$, $D_2 (x_2, y_2, z_2)$, $D_3 (x_3, y_3, z_3)$, $D_4 (x_4, y_4, z_4)$, lengths of four lifting wire ropes are respectively calculated as follows:

a length of a first lifting wire rope of the four lifting wire ropes is

$$C_1 = \sqrt{(x_5 - x_{\textcircled{2}})^2 + (y_5 - y_{\textcircled{2}})^2 + (z_5 - z_{\textcircled{2}})^2},$$

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a length of a second lifting wire rope of the four lifting wire ropes is

$$C_2 = \sqrt{(x_6 - x_2)^2 + (y_6 - y_2)^2 + (z_6 - z_2)^2},$$

a length of a third lifting wire rope of the four lifting wire ropes is

$$C_3 = \sqrt{(x_5 - x_3)^2 + (y_5 - y_3)^2 + (z_5 - z_3)^2},$$

a length of a fourth lifting wire rope of the four lifting wire ropes is

$$C_4 = \sqrt{(x_6 - x_4)^2 + (y_6 - y_4)^2 + (z_6 - z_4)^2},$$

the height (h_4') of the first main hook and the height (h_4'') of the second main hook are calculated by setting the length (C_2) of the second lifting wire rope and the length (C_3) of the third lifting wire rope, so as to obtain the length (C_1) of the first lifting wire rope and the length (C_4) of the fourth lifting wire rope, and then an angle (γ) between each of the four lifting wire ropes and the horizontal plane exceeds 60° , the angle (γ) between each of the four lifting wire ropes and the horizontal plane is calculated as follows:

$$\gamma_1 = \arcsin\left(\frac{z_{\textcircled{2}} - z_{\textcircled{2}}}{C_1}\right), \gamma_2 = \arcsin\left(\frac{z_{\textcircled{2}} - z_{\textcircled{2}}}{C_2}\right),$$

$$\gamma_3 = \arcsin\left(\frac{z_{\textcircled{2}} - z_{\textcircled{2}}}{C_3}\right), \gamma_4 = \arcsin\left(\frac{z_{\textcircled{2}} - z_{\textcircled{2}}}{C_4}\right);$$

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and then a load on each of the four lifting wire rope is calculated as follows: since a gravity of the floating crane is vertically downward and the center of the connecting line between the two main hooks coincides with the overall center of gravity position of the quayside container crane in the top view, a load on the first main hook and a load on the

second main hook is substantially balanced, it is obtained that a vertical downward load on each of the two main hooks is

$$\frac{Q}{2}$$

when a total weight (Q) of the quayside container crane is obtained according to the design drawing, so that the load on the first lifting wire rope, the load on the second lifting wire rope, the load on the third lifting wire rope and the load on the fourth lifting wire rope are respectively calculated with the first lifting wire rope and the third lifting wire rope sharing a first main hook while the second lifting wire rope and the fourth lifting wire rope sharing the second main hook, based on that a resultant force in a vertical direction of two of the four lifting wire ropes of one of the two main hooks is a lifting weight

$$\frac{Q}{2},$$

and a resultant force in a horizontal direction is 0, it is obtained that

$$F_{\textcircled{2}} \times \sin \gamma_{\textcircled{2}} + F_{\textcircled{2}} \times \sin \gamma_{\textcircled{2}} = \frac{Q}{2},$$

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$$F_1 \times \cos \gamma_1 = F_3 \times \cos \gamma_3;$$

$$F_{\textcircled{2}} \times \sin \gamma_{\textcircled{2}} + F_{\textcircled{2}} \times \sin \gamma_{\textcircled{2}} = \frac{Q}{2},$$

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$F_2 \times \cos \gamma_2 = F_4 \times \cos \gamma_4$, the load on the first lifting wire rope, the load on the second lifting wire rope, the load on the third lifting wire rope and the load on the fourth lifting wire rope are obtained by solving equations respectively:

$$F_{\textcircled{2}} = \frac{Q}{2(\sin \gamma_{\textcircled{2}} + \cos \gamma_{\textcircled{2}} \times \tan \gamma_{\textcircled{2}})}, F_{\textcircled{2}} = \frac{Q}{2(\sin \gamma_{\textcircled{2}} + \cos \gamma_{\textcircled{2}} \times \tan \gamma_{\textcircled{2}})},$$

$$F_{\textcircled{2}} = \frac{Q}{2(\sin \gamma_{\textcircled{2}} + \cos \gamma_{\textcircled{2}} \times \tan \gamma_{\textcircled{2}})}, F_{\textcircled{2}} = \frac{Q}{2(\sin \gamma_{\textcircled{2}} + \cos \gamma_{\textcircled{2}} \times \tan \gamma_{\textcircled{2}})}$$

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according to the load on the first lifting wire rope, the load on the second lifting wire rope, the load on the third lifting wire rope and the load on the fourth lifting wire rope, wherein a lifting wire rope having a diameter with

a safety factor above 4 and a shackle with a safety coefficient above 1 are selected by referencing a wire rope breaking force table;

step 6: making preparations before lifting; and

step 7: performing lifting operation at the dock.

2. The full-unit lifting method for the quayside container crane according to claim 1, wherein the length of each of the four lifting wire ropes (C_1 - C_4) is 24 m, the angle (Y) between the four lifting wire ropes and the horizontal plane is 60° at minimum, and the height of the main hook to the upper pulley at the boom of the floating crane (h_5) has a minimum value (h_{5min}) of 5 meters.

3. The full-unit lifting method for the quayside container crane according to claim 1, wherein the step 6 further comprises: before lifting, travelling a trolley to a maximum rear extension position of a rear boom, retracting an upper frame of a container hanger to a highest position of the container hanger, parking an elevator at a first floor, locking brakes of the assemblies in such a way, that the trolley and trailer trolley wheels are plugged by wedge blocks, to keep the front boom at a horizontal position; for a travelling mechanism, inserting wooden blocks at rotatable positions of eight-wheel equalizer beams and driving and driven bogies before the lifting, so as to prevent rotation; checking securing and fastening of the assemblies of the quayside container crane to ensure all movable components are fastened; disassembling all anchorages between the quayside container crane and a transport vessel; using two winches on a floating crane vessel to secure two portal legs obliquely opposite the quayside container crane before the lifting, diagonal portal legs of the quayside container crane are pulled by the two winches on the floating crane vessel after the lifting, to maintain a constant inclination angle between the floating crane and the quayside container crane.

4. The full-unit lifting method for the quayside container crane according to claim 1, wherein the step 7 of performing lifting operation at the dock further comprises: 1) after a floating crane vessel arrives at a construction site, the floating crane vessel is positioned within 200 m offshore from a quay front line, the floating crane vessel inclines, a length of an anchor cable is adjusted with reference to barge

parameters and water conditions of the dock, to realize operational processes of lifting, shifting, and parking; 2) after the floating crane vessel is positioned, all tools and equipment used during the lifting are comprehensively checked, the floating crane vessel is put into use after it is confirmed that no abnormal condition exists, and whether a lifting point of the equipment is consistent with a diameter of a shackle pin or not is checked, after checking, an angle of the boom is adjusted to a required angle, all mechanical parts of the floating crane vessel are rechecked, and the floating crane vessel is used after being confirmed to be intact; 3) the floating crane vessel is lifted by the two main hooks, the two main hooks are loosened above the equipment to be lifted, rigging suspended on the two main hooks in advance is attached to lifting lugs of the equipment in sequence, and reliability of each connection point is rechecked; 4) after preparation, the lifting is performed, when the lifting is 200 mm away, all starting equipment brakes, a component is suspended on the two main hooks for secondary braking, a static state is maintained, when there is no abnormal condition, the floating crane is started to lift the component to a preset height and is suspended in a static state, the component is horizontally lifted during the lifting, to avoid uneven height discrepancies between the two main hooks; 5) after the component is lifted, a towing belt of a tugboat is used to assist in shifting to an installation berth, and anchor positioning is performed at an installation site, the floating crane vessel is repositioned after arriving at the installation site by adjusting a length and an orientation of the anchor cable, so that the floating crane is positioned above a rail of an installation dock, and the floating crane is kept descending horizontally, so that the equipment in a suspended state is aligned to a position right above a placing point by 10 cm; 6) a placing position of the equipment is approved, the floating crane is commanded to loosen the two main hooks, and the equipment is placed at the placing point; 7) lifting construction is carried out until the lifting is finished, and all lifting tools are retracted and the floating crane vessel is withdrawn.

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